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Shock tube study of the decomposition of cyclopentyl radicals

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Abstract

The kinetics of the thermal decomposition of cyclopentyl radicals has been studied in a single-pulse shock tube over a temperature range of 950–1116 K and pressures of 2.3–3.4 bar. Cyclopentyl radicals were generated by abstraction of H from cyclopentane by H atoms, which were produced by the thermal decomposition of hexamethylethane. Products were analyzed using dual-column gas chromatography under cryogenic conditions with simultaneous flame-ionization and mass-spectrometric detection. The main stable products resulting from the decomposition of cyclopentyl radicals are ethene and cyclopentene. These results are consistent with the interconversion of cyclopentyl and pent-4-en-1-yl radicals. The product branching ratio of ethene to cyclopentene is determined as [ethene]/[cyclopentene] $= (8.2 \pm 0.7)\exp[-(1067 \pm 88)/T)$, where the listed uncertainty (1σ) is from the statistical analysis. The overall accuracy in our measurement of the branching ratio is estimated at about 8% (1σ). In contrast to previous estimates, ring opening of cyclopentyl is found to be faster than ejection of a hydrogen atom. The former reaction is favored by about a factor of three. An RRKM/Master Equation treatment was adopted to assign the rate parameters to elementary reactions. Details of the model and its compatibility with the present and literature results are presented. Kinetic results for the pyrolysis of cyclopentyl and pent-4-en-1-yl radicals have been derived over relevant combustion conditions [0.1–1000 bar, 700–1900 K].
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Keywords: Cyclopentyl; Pentenyl; Decomposition; Kinetics; Pyrolysis

1. Introduction

Liquid transportation fuels (diesel, gasoline, aviation) contain large quantities of saturated hydrocarbons having straight, branched, and cyclic structures. Alkyl radicals are the main initial species formed during fuel breakdown and their unimolecular decomposition reactions compete with oxidation steps. The pyrolytic branch leads to the formation of smaller fuel molecules and precursors of soot and polycyclic aromatic hydrocarbons (PAH). It involves isomerization of the radicals and degradation via beta-bond scission to smaller unsaturated fragments. 1-Olefinyl radicals are important intermediates. An understanding of the reaction mechanisms, together with reliable kinetic data, are required input data in kinetics models needed for the design of more efficient engines

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and to reduce formation of soot particulates which are environmental pollutants [1–3].

In previous studies we have examined pyrolysis reactions of a range of straight and branched alkyl radicals [4–7]. In the present work we turn to cyclic systems and consider the mechanism and rate constants for the decomposition of cyclopentyl radical. The pertinent reactions are shown in Fig. 1. In general, cyclic radicals can undergo reversible isomerization to produce 1-olefinyl radicals in competition with C–H bond scission that produces cycloalkenes. These latter species may result in PAH formation pathways and are therefore of interest. In combustion systems the cyclic radical can be formed either directly by abstraction of H from the cycloalkane or via ring closure of the appropriate 1-olefinyl species formed during the breakdown of larger fuel components.

To our knowledge there is no direct experimental study of the thermal gas phase decomposition of cyclopentyl radical. Most of the work reported in the literature [8–14] was carried out either at low temperatures using chemical activation techniques [11–14] or are theoretical studies [8,9]. The most direct information comes from the study of Tsang and Walker [10] on the decomposition of pent-4-en-1-yl. In that work very small amounts of cyclopentene were observed and were attributed to ring closure to cyclopentyl radical and subsequent ejection of a hydrogen atom. Tsang has recently [15] analyzed the existing data from that work in conjunction with the low temperature chemical activation studies of Stein and Rabinovich [13] and Carter and Tardy [11,12] and was able to derive estimates of the rate constants for the system. In the present study we enter the C_5H_9 system from the cyclopentyl side of the reaction. In combination with the previous work this should constrain the results to a much greater degree.

2. Experimental procedures [16]

Experiments have been carried out in a heated single pulse shock tube. Details of the instrument and procedures can be found in earlier publications [17–19]. The experimental temperatures ranged from 950 to 1116 K with shock pressures of 2.3–3.4 bar. Heating times, as derived from recorded pressure traces, were about 500 μs.

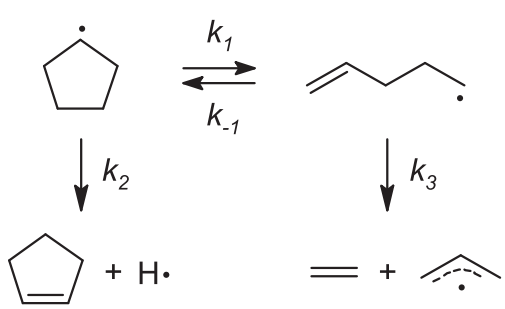

Fig. 1. Mechanism for the thermal decomposition of cyclopentyl radicals.

Cyclopentane is thermally stable under our conditions [20]. To generate cyclopentyl radicals, we place a small amount (typically 100 μL/L) of a hydrogen atom precursor in the presence of a large excess (20,000 μL/L) of cyclopentane. Hydrogen atoms react rapidly with the excess cyclopentane via:

$$\text{cyclopentane} + \text{H} \rightarrow \text{cyclopentyl} + \text{H}_2 \quad (1)$$

We form the necessary hydrogen atoms by the thermal decomposition of hexamethylethane (equations 2 and 3), a technique used in this laboratory for many years [21].

$$\text{hexamethylethane} \rightarrow \textit{tert}\text{-butyl} + \textit{tert}\text{-butyl} \quad (2)$$

$$\textit{tert}\text{-butyl} \rightarrow \text{isobutene} + \text{H} \quad (3)$$

Chain processes are controlled by the addition of about 6500 μL/L of the free radical scavenger 1,3,5-trimethylbenzene (135TMB). Some H atoms are lost by reaction with 135TMB, either via abstraction of H to give the unreactive 3,5-dimethylbenzyl radical, or displacement of methyl from the ring to produce *meta*-xylene. Methyl radicals so produced can similarly abstract H from either 135TMB or cyclopentane; in the latter case they are simply an alternate source of the desired cyclopentyl radicals.

Product analysis employs an Agilent Technologies 6890N gas chromatograph (GC) equipped with a J&W Scientific DB-1 30 m × 0.53 mm internal diameter (i.d.) fused silica capillary column for the larger species and a Restek 30 m × 0.53 mm i.d. Rt-Alumina (aluminum oxide porous layer) capillary column for the lighter components. The GC is equipped with both flame ionization detection (FID) and an Agilent Technologies 5973 Inert mass selective detector. The sample eluting from the DB-1 column was split for simultaneous identification and quantification of the mixture components by mass spectrometry and FID analysis. The GC oven temperature was programmed from −60 to 180 °C.

Shock temperatures are determined by following the progression of a standard reaction with a known rate constant. The present study utilized the reverse Diels-Alder reaction of 4-vinylcyclohexene which has the known [17] rate expression k(4-vinylcyclohexene → 1,3-butadiene + 1,3-butadiene) = $10^{15.4}\exp(-31{,}100/T)\ \text{s}^{-1}$. Shock pressures are calculated from the temperature and mixture composition via the ideal shock equations [22].

Most experiments were conducted with mixtures composed of 100 μL/L hexamethylethane, 20,000 μL/L cyclopentane, 100 μL/L 4-vinylcyclo-

hexene and 6500 μL/L 135TMB in high-purity Argon (Praxair, 99.999%). A few experiments were carried out without added 135TMB, with no change in the product ratio of interest. Cyclopentane (99.9%, Aldrich), hexamethylethane (98%, Aldrich), 4-vinylcyclohexene and 135TMB (99%, Aldrich) were used without further purification other than degassing during preparation of the mixtures.

FID responses of all the olefins were determined from standard samples. Peak areas were determined using the HP ChemStation software with conversion to molar quantities based on the experimentally determined response factors.

3. Results and discussion

3.1. Main products and mechanism

Product data from individual experiments are provided in Supplementary Table S1. Ethene, isobutene, 1,3-butadiene, and cyclopentene are the major olefinic products. Isobutene stems from our H atom source, and 1,3-butadiene from our temperature standard. The other olefins are attributed to decomposition of cyclopentyl radical. Once formed, cyclopentyl radicals undergo a reversible isomerization reaction to give pent-4-en-1-yl radicals in competition with C–H bond fission to produce cyclopentene and H atom (Fig. 1). The main reaction channel for pent-4-en-1-yl radical involves breaking of the allylic C–C bond to form allyl radical and ethene. Since the alkene products are stable under our conditions and there is no other source for their formation, the yields of ethene and cyclopentene directly correspond to the cracking pattern for cyclopentyl radical. Under steady state conditions one obtains:

$$\frac{[\text{ethene}]}{[\text{cyclopentene}]} = \frac{k_1 k_3}{k_2(k_{-1} + k_3)} \tag{4}$$

The ratio is equal to the relative rate of ring-opening to C–H bond fission (k_1/k_2) in the limiting case where $k_3 \gg k_{-1}$. The experimental data are plotted as a function of temperature in Fig. 2 and may be represented as:

$$[\text{ethene}]/[\text{cyclopentene}] = (8.2 \pm 0.7)\exp[-(1066 \pm 88)/T],$$

where the listed uncertainty (1σ) is precision only.

The only other isomerization reaction of potential interest in this system involves a *1,3* H transfer:

$$\text{pent-4-en-1-yl} \rightarrow \text{pent-1-en-3-yl} \tag{5}$$

Our previous experimental work has shown that *1,3* H transfers are generally slow relative to β C–C bond scission, although process (5) could be promoted by the allylic hydrogen. Hayes and Burgess [23] calculate that the allylic effect lowers the abstraction barrier by about 12 kJ mol^{-1} but the barrier remains near 150 kJ mol^{-1}. This is almost a factor of two larger than that expected [9,24] for fission of the allylic C–C bond in pent-4-en-1-yl. If pent-1-en-3-yl were present one might expect formation of 1,3-butadiene via (*pent-1-en-3-yl → 1,3-butadiene + CH_3*). We did find traces of 1,3-butadiene in a few experiments carried out in the absence of the temperature standard 4-vinylcyclohexene (which itself forms 1,3-butadiene as a product), but the molar amount corresponded to only about 0.4% of ethene, which confirms that (5) is negligible under our conditions.

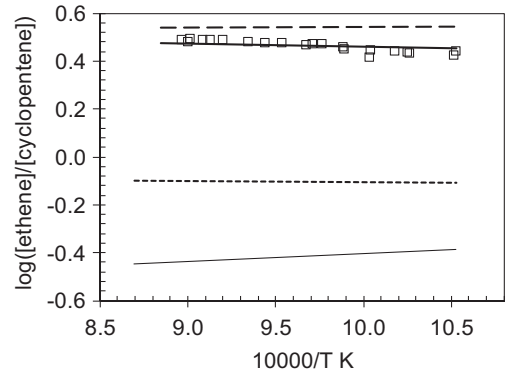

Fig. 2. Ethene to cyclopentene yields as a function of temperature from the decomposition of cyclopentyl radicals. The squares are our experimental data. The upper solid line is the fit from the RRKM/Master Equation model described in the text. The other lines represent the high pressure k_1/k_2, ratios from the present work (upper dashed line), the fit of Tsang [15] (middle dotted line) and the CBS-QB3 calculations of Sirjean et al. [9] (lower solid line).

3.1.1. Secondary chemistry

Secondary chemistry is minimal in this system. Tsang [20] has investigated the thermolysis of cylcopentane and reported channels leading to *1-pentene* and *cyclopropane + ethene*. Consistent with his observations, traces of 1-pentene and cyclopropane were noted, but the pyrolytic channels are overwhelmed in the present system by decomposition induced by hydrogen atoms. The kinetics and observed product amounts dictate that the maximum possible corrections to our ethene yields due to pyrolysis reactions of cyclopentane of cyclopentane are a few percent or less.

With respect to radical-induced decomposition, product data (Table S1) indicate short chain lengths with formation of only about 1.3 cyclopentyl radicals per H atom formed from hexamethylethane decomposition. The allyl radicals formed in the decomposition of pent-4-en-1-yl are largely stable on the time scale of our experi-

ment and act as inhibitors in a manner similar to the benzylic radicals from 135TMB. Their presence is confirmed by the observation of small amounts of the expected radical-radical combination products. These include 1,5-hexadiene from the *allyl + allyl* reaction, and the C12 species formed from *allyl + 3,5-dimethylbenzyl*. The latter species arises from the 135TMB inhibitor as discussed earlier. The recombination *3,5-dimethylbenzyl + 3,5-dimethylbenzyl* must also occur, but the C18 product does not elute from the GC column under our analytical conditions. Small amounts of other products are also observed: these are, in order of decreasing molar production, methane, propene, allene, cyclopentadiene, and ethane. These are accounted for by side reactions:

$$CH_3 + RH \rightarrow CH_4 + R \quad (6)$$

$$CH_3 + CH_3 \rightarrow C_2H_6 \quad (7)$$

$$C_3H_5(\text{allyl}) \rightarrow \text{allene} + H \quad (8)$$

$$C_3H_5 + RH/H \rightarrow \text{propene} + R \quad (9)$$

$$C_3H_5 + C_3H_5 \rightarrow 1,5\text{-hexadiene} \quad (10)$$

$$C_3H_5 + 3,5\text{-dimethylbenzyl} \rightarrow C_{12}H_{16}[1\text{-(3-butenyl)-3,5-dimethylbenzene}] \quad (11)$$

$$\text{Cyclopentene} + R \rightarrow \text{cyclopentenyl}(+RH) \rightarrow \text{cyclopentadiene} + H \quad (12)$$

Small amounts of 1-butene from $C_3H_5 + CH_3$ are also expected but this product co-elutes with cyclopentane on the alumina column and with isobutene on the DB-1 column; peak areas for the *isobutene + 1-butene* peak were consistent with isobutene expected based on loss of hexamethylethane, so only small amounts of 1-butene could be present. Note that the secondary products are largely limited to reactions involving highly stabilized radicals. With respect to secondary reactions involving the main olefin products, ethene has no weak C–H bonds, and the major chemistry of concern would be radical attack on cyclopentene, which leads easily to cyclopentadiene (equation 12). Such reactions are largely precluded by short chain lengths and the overwhelming excess of scavenger and cyclopentane, but any cyclopentadiene formed is derived from cyclopentene and the two should be summed in the determination of the branching ratio for cyclopentyl decomposition. In our GC analyses cyclopentadiene co-eluted with an impurity (2,2-dimethylbutane, present in hexamethylethane), but could be quantified (Table S1) at temperatures $\geqslant 1070$ K on the basis of the integrated $m/e = 66$ ion ($C_5H_6^+$) in the mass spectrum in conjunction with a separately calibrated response factor. At lower temperatures signal levels were too small to obtain accurate results. Between 1070 and 1116 K the correction to the cyclopentene yield increased smoothly from 1.0% to 2.6%. Adjustments at lower temperatures are smaller and were approximated by extrapolating the data from higher temperatures. The minor nature of this correction highlights our ability to avoid contributions from complex secondary chemistry by working with sensitive detectors, efficient radical chain scavengers, and low concentrations of the radical precursor. As noted, all observed minor products are consistent with expectations and none of the secondary chemistry significantly impacts the ethene/cyclopentene ratio, which is the quantity of interest. These reactions are not discussed further in this report.

3.2. Kinetic model

Decomposition of cyclopentyl radical involves reversible opening of the ring via *beta* C–C bond scission and two decomposition channels that produce the observed stable alkene products. Product ratios depend on the algebraic relationship of rate constants given in Section 3.1 and the rates are pressure dependent. The experimental results can thus not be extrapolated to other conditions without the use of a reaction model. We have consequently modeled the decomposition reactions of cyclopentyl radical using a Rice Ramsberger Kassel Marcus/Master Equation (RRKM/ME) analysis to fit the data and allow extrapolation over a range of pressure and temperature conditions of interest to combustion scientists. RRKM/ME calculations were carried out with the ChemRate software package [25,26].

3.2.1. Thermodynamic properties and RRKM/ME treatment

The reaction involves the establishment of a partially equilibrated steady-state concentration of cyclopentyl and pent-4-en-1-yl radicals. The thermodynamic properties of these radicals thus play a role in the determination of the rate constants. Structural properties (geometries, vibrational frequencies, internal and external moments of inertias) for the radicals and stable species are reported in the supplementary data and were based on computations at the G3MP2B3 level of theory. Frequencies from the calculations have been scaled by 0.96 and those corresponding to internal rotational modes have been removed and approximated with symmetric *n*-fold potentials using the methodology of Pitzer [27]. For the stable compounds the derived thermodynamic parameters were in close agreement with literature values [28–30].

Transition state structures for the reactions were estimated in two ways. In the first method they were derived from the appropriate parent species by removing the frequency corresponding to the stretch mode that becomes the reaction coordinate. Thus the ring opening reaction (k_1) was based on cyclopentyl while the C–C fission

yielding ethene and allyl (k_3) was based on pent-4-en-1-yl. Barriers for internal rotors in the transition state were not changed except for those corresponding to the incipient double bond and that of the breaking bond, which were frozen and allowed to become free, respectively. Alternatively, structures were derived from calculations at the G3MP2B3 level of theory. These data are listed in the supporting information and will be reported in more detail elsewhere. In either case slight adjustments are made to the low frequency vibrations to create a model that fits the experimental data. Both methodologies gave acceptable results in terms of our ability to reproduce the data using reasonable rate expressions. For consistency our final fits are based on the theoretical structures.

Enthalpies of formation of cyclopentene and ethene were taken from the evaluation of Frenkel [31]. For cyclopentyl and allyl radicals we have used the values of Castelhano and Griller [32], and Tsang [33], respectively. For pent-4-en-1-yl we have derived a value based on a bond dissociation energy of 420 kJ mol^{-1} for the methyl C–H bond in 1-pentene, and taken ΔH_f(1-pentene, 298 K) = −21 kJ mol^{-1}. This latter value is based on our evaluation of the available experimental values [34–36], ranging from −17 to −26 kJ mol^{-1}, and is close to that derived from group additivity. Agreement with values derived from the G3MP2B3 calculations was typically ±6 kJ mol^{-1}.

Equilibrium constants of formation for the species in the model are presented in Table 1 together with the final high pressure limiting rate expressions. Because the model fit depends on the balance of forward and reverse reaction rates the use of alternate thermodynamic properties would require compensating changes in the kinetic parameters to maintain the fit to the experimental branching ratios. This should be borne in mind if the present results are incorporated into existing combustion models.

Our RRKM/ME analysis involves a time dependent treatment in which we assume that there is a constant incoming flux of the radical of interest (cyclopentyl or pent-4-en-1-yl) having a Boltzmann energy distribution corresponding to the reaction temperature. The energy distributions of the reacting molecules then evolve based on the microcanonical reaction rates and collisional energy transfer during the period of reaction until the traditional steady state solution is reached. The reader is referred to our previous publications for details [4,7,19]. All experiments in the present study were performed in argon, and collisional energy transfer has been treated with an exponential-down model [37] with $\alpha = 0.293T$ K^{-1} cm^{-1}, where α is the usual parameter that relates to the average energy transferred in deactivating collisions, and T is the temperature in kelvin. Tunneling contributions are expected to be small and have not been included.

3.2.2. Fitting procedures

The experimental yields of ethene and cyclopentene provide information on the relative rate constants for the processes shown in Fig. 1. To place the results on an absolute basis the rate expression for at least one reaction must be fixed. We opted to fix the rate constant for H atom addition to cyclopentene, which we are presently studying in separate experiments. The parameters for this reaction over the temperature range of 1027–1167 K were taken from our preliminary results that will be reported elsewhere. In that study the rate of H atom addition to cyclopentene was determined relative to displacement of methyl radical from 135TMB. From the literature value $k[H + (CH_3)_3C_6H_3 \rightarrow (CH_3)_2C_6H_4 + CH_3] = 6.7 \times 10^{13}\ \exp(-3255/T)$ cm^3 mol^{-1} s^{-1} [38], which is ultimately linked to the well-known rate constant for $H + CH_4$, we find k^{∞}(H + cyclopentene → cyclopentyl) = $7.69 \times 10^{13}\ \exp(-1626/T)$ cm^3 mol^{-1} s^{-1}. Near 1000 K the standard uncertainty in the absolute value of this rate constant is estimated to be ±50%. Detailed balance leads to a high pressure rate expression for k_2, the ejection of a hydrogen atom from cyclopentyl radical.

With k_2 fixed, the initial barriers for the other reactions were based on the G3MP2B3 calculations for the transition states, other recent computational estimates [9,24,39], together with consideration of older literature estimates derived from the low temperature chemical activation studies [11–14]. Variations in the available barrier heights were in the range of 8 to 20 kJ mol^{-1} (see Table S3 in the supporting information for some comparisons with computed values). Adopted parameters were used to calculate the steady-state concentrations of the radicals at the reaction temperatures and pressures using our RRKM/ME treatment and combined with the rate constants of the product forming reactions to obtain product concentrations. None of the previously proposed parameter sets were able to simultaneously reproduce both the shock tube results of Tsang and Walker on the decomposition of pent-4-en-1-yl radical and the present data on cyclopentyl radical. Using an iterative procedure, small adjustments were then made to the non-fixed rate constants so as to match the experimental ethene to cyclopentene ratios in both studies. Necessary adjustments to the rate constants were made either by altering the enthalpies of formation of the transition states or by modifying their low frequency modes to adjust the pre-exponential factors.

To help guide the fits, the cyclopentyl/pent-4-en-1-yl system was modeled using the plug flow reactor module of CHEMKIN-PRO [40] and a linear sensitivity analysis performed with this software. Plots of the normalized sensitivity coefficients (NSC) from the final fit are shown in Fig. 3. It is found that when the starting radical is cyclopentyl the alkene product ratio is primarily

Table 1
Kinetic [A, B] and thermodynamic [C] properties of cyclopentyl radical decomposition.

k	Reaction	Log A	n	E/R (K)	Log $k\infty$ (1000 K)
A: high pressure limiting rate expressions[a]: k^{∞}(700–1900 K) = A(T/298 K)^{n}exp(-E/T) s^{-1}					
1	cyclopentyl → pent-4-en-1-yl	14.26	0.407	17506	6.87
−1	pent-4-en-1-yl → cyclopentyl	11.04	0.893	7733	8.15
2	cyclopentyl → cyclopentene + H	13.34	0.821	17138	6.32
3	pent-4-en-1-yl → ethene + allyl	12.79	0.73	9535	9.03
B: addition reactions (from detailed balance): k^{∞}(700–1900 K) = A (T/298 K)^{n}exp(-E/T) cm^{3} mol^{-1} s^{-1}					
−2	cyclopentene + H → cyclopentyl	12.37	1.47	−91	13.18
−3	ethene + allyl → pent-4-en-1-yl	9.23	2.57	4627	8.57

C: thermodynamic properties (ΔH_f in kJ mol^{-1}; S in J mol^{-1} K^{-1})

Compound	$\Delta H_{f,298}$	S_{298}	Equilibrium constant of formation, log [K_p]
Cyclopentyl	110	306.9	$-38.927 + 2.43\ln T - 2.16 \times 10^{-4}T + 263T^{-1} - 1.474 \times 10^{6}T^{-2} + 1.660 \times 10^{8}T^{-3} - 7.10 \times 10^{9}T^{-4}$
Pent-4-en-1-yl	185	352.4	$-30.326 + 1.725\ln T - 2.12 \times 10^{-4}T - 5030T^{-1} - 1.166 \times 10^{6}T^{-2} + 1.337 \times 10^{8}T^{-3} - 5.79 \times 10^{9}T^{-4}$
Cyclopentene	52	231	$-32.901 + 2.027\ln T - 2.000 \times 10^{-4}T + 3579T^{-1} - 1.320 \times 10^{6}T^{-2} + 1.490 \times 10^{8}T^{-3} - 6.39 \times 10^{9}T^{-4}$
Ethene	171	264.8	$-7.688 + 0.4821\ln T - 7.880 \times 10^{-5}T - 653T^{-1} - 5.580 \times 10^{5}T^{-2} + 6.600 \times 10^{7}T^{-3} - 2.90 \times 10^{9}T^{-4}$
Allyl	33.9	292.5	$-12.464 + 0.853\ln T - 1.130 \times 10^{-4}T - 6486T^{-1} - 6.271 \times 10^{5}T^{-2} + 7.270 \times 10^{7}T^{-3} - 3.18 \times 10^{9}T^{-4}$

[a] Rate parameters have been derived so as to fit the data in Figs. 2 and 4. Significant figures are given so as to accurately represent the model results and do not imply accuracy. See Section 4 of text for discussion of uncertainties.

sensitive to the ratio k_1/k_2, while when the starting radical is pent-4-en-1-yl the system shows a similar degree of sensitivity to all rate constants.

3.2.3. Comparisons of the model with experiment and literature results

Figure 2 shows the experimental data from the present work together with the present fit and earlier estimates reported in the literature. Figure 4 compares the experimental ethene to cyclopentene ratio obtained by Tsang and Walker in the decomposition of pent-4-en-1-yl radical with values predicted by the current model. The present model was also tested against the low temperature chemical activation results [11–13], which involve experiments carried out in a hydrogen bath gas. As with the previous analysis by Tsang [15], reproducing these data with our model required us to use similarly low values of the energy transfer parameter. It is not certain if this is indicative only

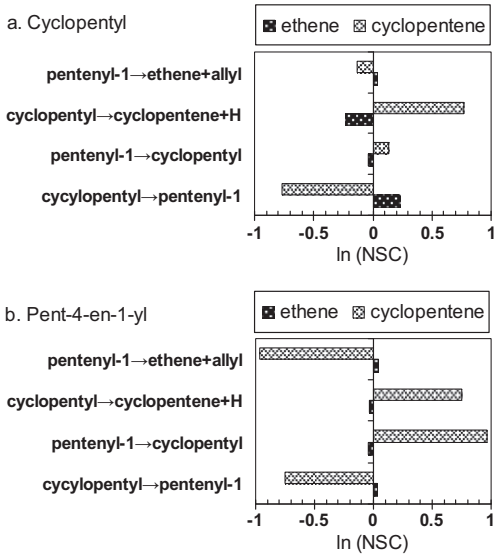

Fig. 3. Normalized Sensitivity Coefficients (NSC) for our pyrolysis model when the starting radical is cyclopentyl (a) or pent-4-en-1-yl (b).

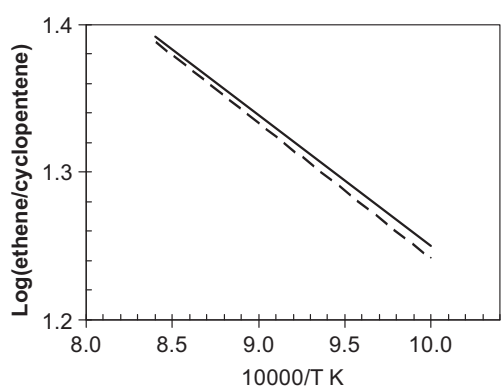

Fig. 4. Ethene to cyclopentene yields from the decomposition of pent-4-en-1-yl radical. The dashed line represents the experimental results of Tsang and Walker [10]. The solid line gives the results of our RRKM/ME model when pent-4-en-1-yl is the starting radical.

of the known low collisional efficiency of H_2, or if it has implications for theory or the accuracy of the potential energy surface of the present reaction.

The potential energy surface used in our model is shown in Fig. 5. The sensitivity of the olefin product ratio from cyclopentyl radical is heavily weighted to k_1/k_2 (Fig. 3), mostly because ring opening is only slightly reversible. The present model has near-equal barriers for fission of the C–H bond in cyclopentyl and its ring opening to give pent-4-en-1-yl (reactions 1 and 2). This is in agreement with the published computational results [9,39]. The similarity of these two barriers is also required to reproduce the experimentally observed weak temperature dependence of the ethene/cyclopentene yields when the starting radical is cyclopentyl (Fig. 2). A consequence is that reproduction of the observed preference for ring-opening requires that this reaction have a pre-exponential factor larger than that of C–H bond fission. This is somewhat surprising as it has been generally thought [14] that the transition state for ring opening should be "tight" compared with that for beta C–H fission. The branching ratio could be matched by simultaneously lowering the barrier for k_1 together with its pre-exponential factor, but this results in a poorer fit to the temperature dependence of the cyclopentyl data. Since the temperature dependence is a key parameter, and could be perturbed by small systematic errors, it would be desirable to have results over a larger temperature range.

The present model uses slightly different pre-exponential factors and a different potential energy surface than the model of Tsang [15]. Our barrier for beta C–C bond scission in pent-4-en-1-yl to give ethene and allyl radical (reaction 3) is about 20 kJ mol^{-1} smaller than Tsang's estimate. At 1000 K the derived rate of this reaction is about a factor of 100 faster than those we have found [4,6,7] for the corresponding scission of a methyl group. This is in reasonable agreement with the Evans–Polanyi analysis of Sayes et al.

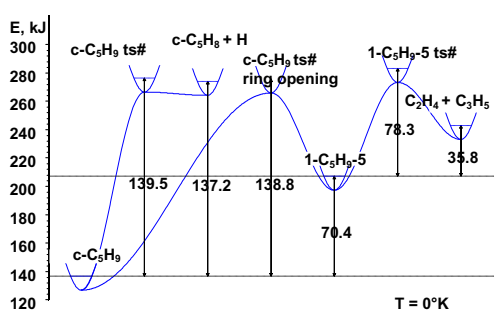

Fig. 5. Potential energy surface from our RRKM/ME model for decomposition of cyclopentyl and pent-4-en-1-yl radicals.

[24]. In terms of the reverse addition of allyl to ethene, our estimate has a smaller activation energy than that proposed by Tsang [15] and our rate constant at 1000 K is about a factor of three faster. Rate differences are larger at lower temperatures, which could have some practical implications for bond formation in reacting systems. There are unfortunately no direct experimental data with which to compare.

Reported computational barriers for C–C bond scission in pent-4-en-1-yl (Supplementary Table S3) show significant variation: at 1000 K the activation energy, $E_{a,3}$, estimated by Saeys et al. [24] on the basis of a CBS-QB3-derived group additivity scheme is 87.5 kJ mol^{-1}, close to our value $E_{a,3,\infty}(1000\ \mathrm{K}) = 85.3$ kJ mol^{-1}. Sirjean et al. [9] derived $E_3(298\ \mathrm{K}) = 84.0$ kJ mol^{-1} from CBS-QB3 calculations compared with our value $E_3(298\ \mathrm{K}) = 80.5$ kJ mol^{-1}. On the other hand, the 0 K barrier of Goldsmith et al. [39] from G3 calculations is 9 kJ mol^{-1} higher than our value and our own G3MP2B3 calculations give a barrier about 17 kJ mol^{-1} higher than the present fit to the experimental data. Our model is in better agreement with the CBS-Q energies which, unlike those from Gn methods, include a significant empirical correction to the energy for spin contamination due to the allylic character of the transition state. [See supplementary data showing a reduction in barrier and heat of reaction by about 10 and 5 kJ mol^{-1}, respectively, due to this empirical correction].

Sirjean et al. [9] have used CBS-QB3 calculations to derive high pressure rate constants for k_1 and k_2. The k_1/k_2 ratio corresponds to the ethene to cyclopentene product ratio from cyclopentyl radical if ring opening is irreversible (Eq. (4)); it therefore represents the maximum value. The value from Sirjean et al. [9] is a factor of 9 lower than indicated by our experimental results at temperatures of 950–1150 K (Fig. 2). The difference is not due to a pressure effect since an extrapolation of the experimental results to the high pressure limit (Fig. S1) only slightly changes the ratio and actually increases the discrepancy. Much of the difference in the branching ratio is attributable to a slower rate of C–H bond scission used by us in comparison with that of Sirjean et al. The discrepancy between the computed and experimental branching ratios highlights the difficulties in precisely predicting channels that occur with similar rates, and demonstrates the need for benchmark experimental data.

4. Conclusions and implications for combustion modeling

The present direct study shows that ring opening of cyclopentyl radical is faster than ejection of hydrogen by about a factor of three near 1050 K.

This compares with the previous estimate by Tsang of approximately equal rates and recent theoretical predictions which suggest that ring opening is a factor of three *slower* than ejection of H atom. This branching ratio may have significant consequences in terms of soot modeling since it suggests that saturated C5 ring structures are less susceptible to thermal dehydrogenation than previously thought; such dehydrogenation processes provide routes to known soot precursors such as cyclopentadiene and cyclopentadienyl radicals.

High pressure rate expressions for pyrolysis of cyclopentyl radical are presented in Table 1. For purposes of extrapolation over combustion conditions, Table S4 presents the departure of the rate constants from their high pressure limits at selected pressures ranging from 0.1 to 1000 bar and temperatures between 700 and 1900 K. Table S5 provides related results for pyrolysis of pent-4-en-1-yl radical. Departures from the high pressure limits are slightly different in the fall-off region for the two cases because the energy distributions evolve differently depending on the starting radical. Results are given in tabular form to allow accurate reproduction over the full range of conditions; for more limited applications the results may be parameterized in a variety of analytical formats if desired.

For many combustion applications the decomposition of the cyclopentyl radical is so fast relative to other processes that the product branching ratio is the important quantity rather than the exact rate constants. The derived temperature dependences of the [ethene]/[cyclopentene] product branching ratios at selected pressures are shown in Figs. S1 and S2 and parameterized in Table S6 of the supplementary data. Over the range of conditions considered this quantity is calculated to vary by about 50% when the starting radical is cyclopentyl and by a factor of ten (3× at a given temperature) for pent-4-en-1-yl. Near the mid-point of our studies the standard uncertainty (1σ) in the experimental product ratio from cyclopentyl radical is estimated at 8% and the value is within about 10% of the high pressure limiting value. At the extremes of our temperature extrapolations, 700 and 1900 K, the uncertainty in the product ratio increases to about 50%. Absolute rate constants become important at lower temperatures where radical lifetimes are longer and bimolecular processes (e.g. oxidation steps) can compete with decomposition. The rate constants derived from the RRKM/ME analysis depend on many parameters in complex ways, making a statistical assessment of uncertainties in the rate constants difficult. Based on empirical observations of the effects of parameter variation, a thermochemical kinetic analysis, and comparisons with related systems, we estimate the standard uncertainties in the absolute rate constants of Table 1 are a factor of about 1.5 near 1000 K, increasing to a factor of 2 at 700 K and a factor of 3 at 1900 K. Note, however, that it is not appropriate to arbitrarily vary the rate constants within these bounds with no regard to the effect on the product branching ratio, which is the quantity defined with greater accuracy.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.109.

Supporting information includes representative chromatograms, product data from individual experiments, Tables of the deviations in k_1, k_{-1}, k_2, and k_3 from high pressure limiting behavior at selected pressures from 700 to 1900 K, molecular properties used in the RRKM/ME calculations, and calculated product branching ratios at selected pressures from 700 to 1900 K.

References

[1] M. Colket, T. Edwards, S. Williams, et al., in: 45th AIAA Aerospace Sciences Meeting, Paper AIAA-2007-770, Reno, NV, 2007.
[2] J.T. Farrell, N.P. Cernansky, F.L. Dryer, et al., in: 2007 SAE World Congress, Paper SAE 2007-01-0201, Detroit, MI, 2007.
[3] H. Richter, J.B. Howard, *Prog. Energy Combust.* 4 (2000) 565–608.
[4] W. Tsang, J.A. Walker, J.A. Manion, *Proc. Combust. Inst.* 31 (2007) 141–148.
[5] W. Tsang, J.A. Walker, J.A. Manion, *Proc. Combust. Inst.* 27 (1998) 135–142.
[6] W. Tsang, W.S. McGivern, J.A. Manion, *Proc. Combust. Inst.* 32 (2009) 131–138.
[7] W.S. McGivern, I.A. Awan, W. Tsang, J.A. Manion, *J. Phys. Chem. A* 112 (30) (2008) 6908–6917.
[8] D.M. Matheu, W.H. Green, J.M. Grenda, *Int. J. Chem. Kinet.* 35 (3) (2003) 95–119.
[9] B. Sirjean, P.A. Glaude, M.F. Ruiz-Lopez, R. Fournet, *J. Phys. Chem. A* 112 (46) (2008) 11598–11610.
[10] W. Tsang, J.A. Walker, *J. Phys. Chem.* 96 (21) (1992) 8378–8384.
[11] W.P.L. Carter, D.C. Tardy, *J. Phys. Chem.* 78 (22) (1974) 2201–2211.
[12] W.P.L. Carter, D.C. Tardy, *J. Phys. Chem.* 78 (16) (1974) 1573–1578.
[13] S.E. Stein, B.S. Rabinovitch, *J. Phys. Chem.* 79 (3) (1975) 191–198.
[14] T. Gierczak, J. Gawlowski, J. Niedzielski, *Int. J. Chem. Kinet.* 18 (6) (1986) 623–637.

[15] W. Tsang, *J. Phys. Chem. A* 110 (27) (2006) 8501–8509.
[16] Disclaimer, Certain commercial materials and equipment are identified in this paper in order to specify adequately the experimental procedure. In no case does such identification imply recommendation of endorsement by the National Institute of Standards and Technology, nor does it imply that the material or equipment is necessarily the best available for the purpose.
[17] W. Tsang, in: Shock Waves in Chemistry, A. Lifshitz (Ed.), Marcel Decker, New York and Basel, 1981, pp. 59–129.
[18] W.S. McGivern, J.A. Manion, W. Tsang, *J. Phys. Chem. A* 110 (47) (2006) 12822–12831.
[19] I.A. Awan, W.S. McGivern, W. Tsang, J.A. Manion, *J. Phys. Chem. A* 114 (30) (2010) 7832–7846.
[20] W. Tsang, *Int. J. Chem. Kinet.* 10 (6) (1978) 599–617.
[21] J.A. Manion, W. Tsang, *J. Phys. Chem.* 100 (17) (1996) 7060–7065.
[22] A.G. Gaydon, I.R. Hurle, The Shock Tube in High-Temperature Chemical Physics, Reinhold Publishing, New York, 1963.
[23] C.J. Hayes, D.R. Burgess Jr., *J. Phys. Chem. A* 113 (2009) 2473–2482.
[24] M. Saeys, M.F. Reyniers, G.B. Marin, V. Van Speybroeck, M. Waroquier, *AIChE J.* 50 (2) (2004) 426–444.
[25] W. Tsang, V. Bedanov, M.R. Zachariah, *J. Phys. Chem. A* 100 (1996) 4011–4018.
[26] V. Mokrushin, W.S. McGivern, V. Bedanov, W. Tsang, M. Zachariah, V.D. Knyazev, ChemRate, Version 1.5.6, National Institute of Standards and Technology, Gaithersburg, Maryland, 2008.
[27] K.S. Pitzer, *J. Chem. Phys.* 12 (1944) 310–314.
[28] D.R. Stull, D.F. Westrum, G.C. Sinke, *The Chemical Thermodynamics of Organic Compounds*, John Wiley and Sons, New York, 1969.
[29] L.V. Gurvich, I.V. Veyts, C.B. Alcock (Eds.), Thermodynamic Properties of Individual Substances, vol. 2, fourth ed., Hemisphere, New York, 1991.
[30] O.V. Dorofeeva, L.V. Gurvich, V.S. Jorish, *J. Phys. Chem. Ref. Data* 15 (2) (1986) 437–464.
[31] M. Frenkel, G.J. Kabo, K.N. Marsh, G.N. Roganov, R.C. Wilhoit, Thermodynamics of Organic Compounds in the Gas State, Thermodynamics Research Center, College Station, TX, 1994.
[32] A.L. Castelhano, D. Griller, *J. Am. Chem. Soc.* 104 (13) (1982) 3655–3659.
[33] W. Tsang, in: Energetics of Organic Free Radicals, J.A. Martinho Simoes, A. Greenberg, J.F. Liebman (Eds.), Blackie Academic and Professional, London, 1996, pp. 22–58.
[34] D.W. Rogers, S. Skanupon, *J. Phys. Chem.* 78 (25) (1974) 2569–2572.
[35] W.D. Good, N.K. Smith, *J. Chem. Thermodyn.* 11 (2) (1979) 111–118.
[36] K.B. Wiberg, D.J. Wasserman, E. Martin, *J. Phys. Chem.* 88 (16) (1984) 3684–3688.
[37] H.H. Carstensen, A.M. Dean, Comprehensive Chemical Kinetics, vol. 42, 2007, pp. 105–187.
[38] W. Tsang, J.P. Cui, J.A. Walker, in: 17th International Symposium on Shock Waves and Shock Tubes, American Institute of Physics, Bethlehem, PA, USA, 1989, pp. 63–73B.
[39] C.F. Goldsmith, H. Ismail, P.R. Abel, W.H. Green, *Proc. Combust. Inst.* 32 (2009) 139–148.
[40] CHEMKIN-PRO, Reaction Design, San Diego, CA USA, 2008.